

Droplet Chromatography (10x Chromium): cell identity is bought with a container, not built into the sequence

Yeast-adapted 10x Chromium of Jariani 2020 Panels read a to c: the container is made, the wall comes off inside it, and a barcoded molecule comes out. Segment colors and the 5'-to-3' convention are those of Fig. 1, so the two methods can be compared directly.

a Co-encapsulation: one cell plus one barcoded bead in one droplet

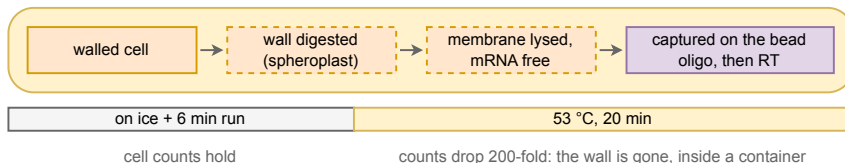
Cells enter the chip **walled and intact** – the wall is what lets them survive the journey.



Loading is Poisson, so occupancy is kept low. You pay per channel of reagent, and cost scales with cells.

b Why the zymolyase rides in the RT mix: the wall comes off only after the cell is already isolated

1 μ L of water in the RT master mix is replaced with 100 $\times$ zymolyase, so digestion, lysis and RT all happen in one droplet, in that order.

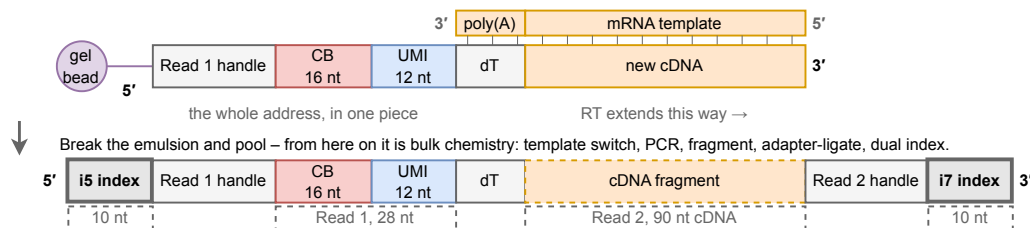


What this avoids

Spheroplast *before* loading and wall-less cells shear in the chip; the mRNA they release contaminates every droplet downstream.

c One bead delivers the whole address at once – compare Fig. 1, where it arrives in three instalments

The 16 nt barcode is pre-synthesised and identifies the *bead*; the 12 nt UMI is random on each oligo copy.



Both families put the barcode on the reverse-transcription primer; what differs is the thing that holds one cell's molecules together. Split-pool uses the fixed cell as the vessel and writes the address over three ligation rounds (Fig. 1); droplet rents an oil-walled container and delivers a 16 nt bead barcode plus a 12 nt UMI in one step, which is why its run needs 118 cycles against 160. The container is also the constraint: it must exist before the wall comes off, and it is paid for per channel.

Segments transcript / cDNA cell barcode UMI gel bead droplet / oil handle / adapter i5 / i7 index || base pairing dashed = disrupted